Ideaverse: Structuring and Steering Design Brainstorming with a Multi-Agent LLM System

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Introduction

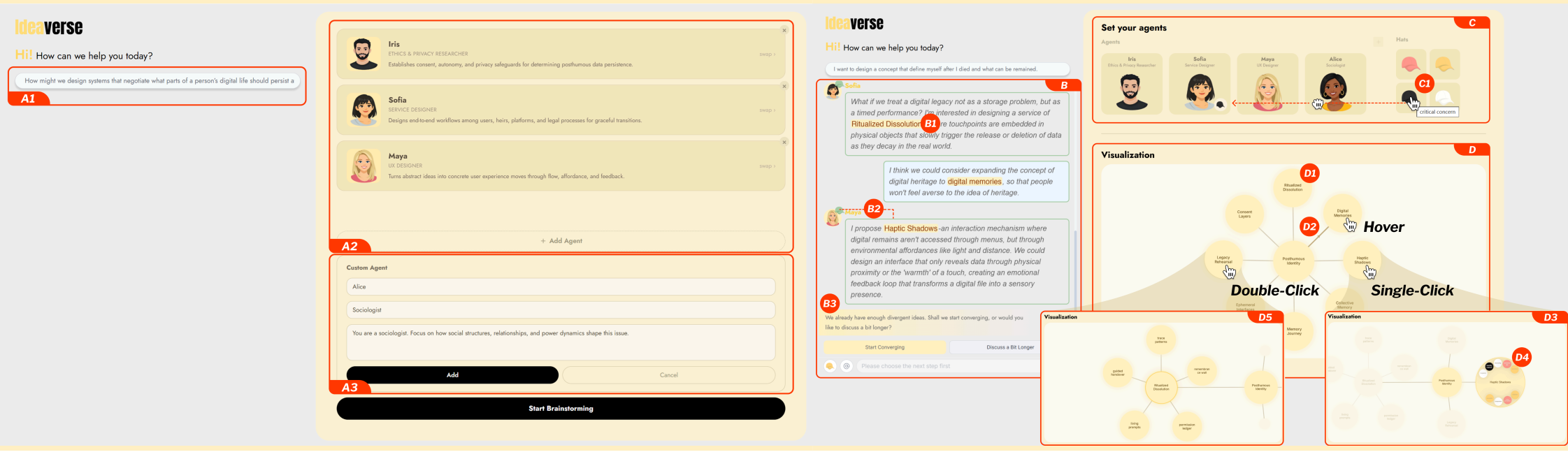
Brainstorming helps designers expand the solution space and develop alternative ideas. Although it is often associated with group workshops, it also commonly happens individually in early-stage design practice. In solo brainstorming, designers need to generate ideas, avoid fixation, organize emerging directions, and decide when to shift from exploration to evaluation on their own.

Gaps

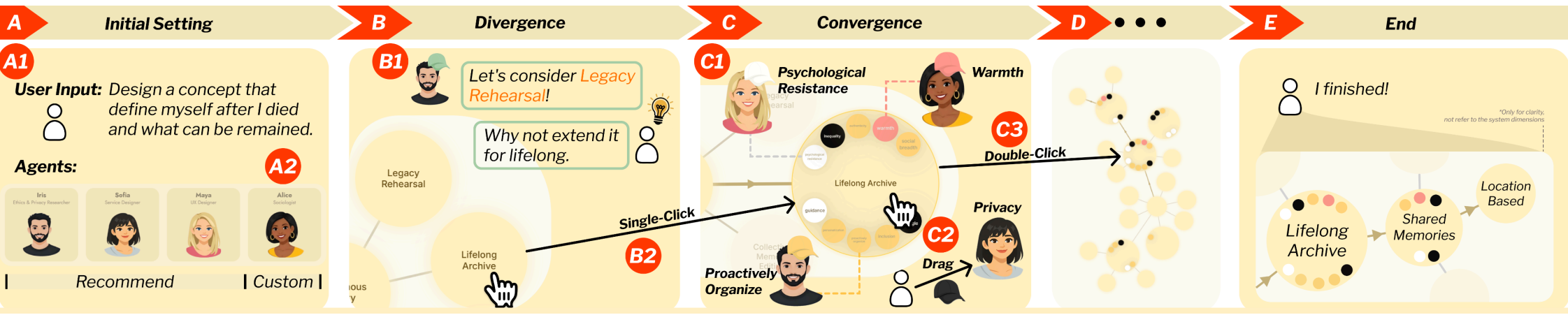
1. Limited cognitive perturbation——Solo designers lack external perspectives, disagreement, and cognitive friction, making ideas more likely to stay close to existing assumptions.
2. High convergence burden——Designers must manually compare, filter, and synthesize many emerging ideas. AI-generated outputs may increase idea quantity, but they are often opaque and difficult to evaluate reflectively.
3. Weak transition support——Many AI-assisted brainstorming systems follow a generate-then-filter workflow, separating divergence and convergence. However, design ideation requires continuous movement between exploration, evaluation, and reframing.

System Design

1. Agent Team Setup
Users enter a topic, choose recommended agents, or create custom roles. (A1–A3)
2. Role × Thinking Hat Mechanism
Agents combine domain roles with thinking hats, generating ideas in divergence and evaluating them from supportive, critical, objective, or emotional perspectives in convergence.
3. Multi-Agent Dialog Interface
Users can observe or intervene during autonomous agent discussion. Keywords, hat icons, and facilitator guidance make key ideas, agent stances, and process cues visible. (B1–B3)
4. Agent Perspective Control
Users can directly assign thinking hats to specific agents, introducing missing perspectives or steering the discussion toward desired evaluative angles. (C1)
5. Idea Graph for Steering
Ideaverse maps ideas into an interactive graph: single-click for local evaluation, double-click for new divergence, with evaluation markers colored by thinking hats. (D1–D5)



System Overview



Usage Scenario

User Study

A between-subjects study (N=18) compared Ideaverse to a baseline.

Comparison across usability and creativity support metrics

Categories	Factors	*Ideaverse* Mean/S.D.	Baseline Mean/S.D.	*p*-value
Foundations	Usability ↑	5.78/0.44	5.22/1.30	0.253
	Learnability ↑	6.55/0.53	6.11/0.93	0.234
	Frustration ↓	2.00/0.70	3.78/1.86	**0.022***
Divergence	Exploration ↑	6.44/0.53	4.78/1.48	**0.009****
	Expressiveness ↑	6.33/1.00	5.44/1.33	0.131
Convergence	Confidence ↑	6.22/0.44	5.11/1.05	**0.014***
	Effort ↓	2.88/0.78	3.78/1.48	0.137
Transition	Ownership ↑	6.44/0.53	5.44/1.13	**0.034***
	Fluidity ↑	6.78/0.44	4.56/2.24	**0.017***
Overall	Creativity ↑	6.67/0.50	5.22/2.22	0.090

Statistical Comparison of User Interaction Metrics

Metrics	*Ideaverse* Mean/S.D.	Baseline Mean/S.D.	*p*-value
Duration (mins)	30.30/7.00	16.88/3.78	< .001 ***
# of Utterances	62.00/16.36	23.33/5.83	< .001 ***
f of Utterances	2.09/0.55	1.40/0.22	0.005 **
# of Agent Utterances	45.78/11.87	13.78/2.54	< .001 ***
f of Agent Utterances	1.55/0.38	0.83/0.09	0.004 **
# of User Utterances	16.22/4.87	9.56/3.47	< .001 ***
f of User Utterances	0.55/0.18	0.57/0.15	0.809

Results of the blind peer evaluation

Factors	*Ideaverse* Mean/S.D.	Baseline Mean/S.D.	*p*-value
Novelty ↑	5.61/0.98	4.02/1.90	< .001 ***
Workability ↑	5.07/1.48	4.53/1.81	0.091
Relevance ↑	5.69/1.18	4.30/1.70	< .001 ***
Specificity ↑	5.76/1.32	3.81/2.06	< .001 ***

Conclusion

Finding 1: Ideaverse shifts AI support from producing more ideas to helping users steer the creative process.
Finding 2: The idea graph externalizes how ideas branch, converge, and evolve, reducing the burden of tracking complex discussions.
Finding 3: Role-based agents and thinking hats introduce diverse, sometimes conflicting perspectives that help users challenge assumptions.
Finding 4: Users remain in control by choosing when to observe, intervene, evaluate a node, or branch into a new direction.

Conclusion: Ideaverse shows that multi-agent LLMs can support solo brainstorming as a steerable cognitive scaffold. Rather than replacing designers' creativity, it helps them externalize, negotiate, and refine an evolving design space.

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